

FARMO AI - SMART FARMING PLATFORM

Enterprise REST API & Microservices Technical Specification

Version 2.4.0
(Enterprise)

Protocol: HTTPS / JSON /
REST

Status: Production
Live

Database: MySQL 8.0+ /
SQLite

Platform Architecture & Base URLs

Component	Base Endpoint URL	Role & Framework
Node.js Express Server	http://localhost:5000/api	Primary Full-Stack REST API & Live MySQL Connection
Spring Boot Microservices	http://localhost:8080/api	Java Spring Data JPA Microservice Cluster
Production Vercel Cloud	https://farmo-ai.vercel.app/api	Cloud Edge Hosting with Local/Offline Sync
Ollama LLM Engine	http://localhost:11434/api	Local AI Neural Inference (Llama-3 / Mistral)

1. System Health & Platform Diagnostics

Heartbeat and diagnostic services providing real-time telemetry on the Express API server, MySQL/SQLite database engine connectivity, and system uptime.

GET /

Platform Welcome & Root Status

Public Access

Returns server branding, deployment mode, and timestamp.

HEADERS

None required

RESPONSE EXAMPLE [HTTP 200]

```
{
  "message": "FARMO AI Backend Server is Running!",
  "version": "2.4.0",
  "status": "active",
  "timestamp": "2026-09-25T12:00:00.000Z"
}
```

GET /api/health

Database & Microservices Health Check

Public Access

Deep health probe verifying MySQL connection pool latency, active database engine, and uptime.

HEADERS

None required

RESPONSE EXAMPLE [HTTP 200]

```
{
  "status": "OK",
  "engine": "mysql",
  "database": "farmo_ai_db",
  "host": "127.0.0.1:3306",
  "uptime_seconds": 14205,
  "timestamp": "2026-09-25T12:00:00.000Z"
}
```

2. Authentication & Identity Services

User registration, credential verification, JWT token issuance, and profile synchronization for both Farmers and Platform Administrators.

POST /api/auth/register

Farmer Account Registration

Public Access

Registers a new agricultural producer. Automatically provisions farmer profile, district assignment, and farm size records in MySQL.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "name": "Suresh Pillai",
  "email": "suresh.pillai@gmail.com",
  "password": "Farmer@123",
  "phone": "+91 94470 12345",
  "district": "Palakkad",
  "crop": "Organic Paddy (Jyothi)",
  "acres": 3.5,
  "soil_type": "Alluvial"
}
```

RESPONSE EXAMPLE [HTTP 201]

```
{
  "success": true,
  "message": "Farmer account registered successfully!",
  "user": {
    "id": "u_1790170999_a1b2",
    "name": "Suresh Pillai",
    "email": "suresh.pillai@gmail.com",
    "role": "farmer",
    "district": "Palakkad",
    "crop": "Organic Paddy (Jyothi)",
    "acres": 3.5
  },
  "token": "jwt_farmer_u_1790170999_a1b2"
}
```

POST /api/auth/login

User & Admin Authentication

Public Access

Validates credentials against MySQL `users` table. Automatically routes between Farmer Portal and Admin Portal according to account role.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "email": "admin@gmail.com",
  "password": "admin@123"
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Login successful",
  "role": "admin",
  "user": {
    "id": "u_admin_default",
    "name": "Admin Administrator",
    "email": "admin@gmail.com",
    "role": "admin",
    "district": "Kerala"
  },
  "token": "jwt_admin_u_admin_default"
}
```

GET

/api/user/profile

Get Authenticated User Profile

Farmer / Admin Access

Fetches full profile metadata, including primary crop, farm land size, soil classification, and profile avatar.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
email	string	Required	User's registered email address

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": {
    "id": "u_1786594685785_3j73",
    "name": "ASHWIN",
    "email": "aswin1210@gmail.com",
    "phone": "9345675687",
    "district": "Wayanad",
    "crop": "rice (basmathi), corn",
    "acres": 5.5,
    "soil_type": "Alluvial",
    "role": "farmer",
    "joined": "2026-08-13"
  }
}
```

5

PUT /api/user/profile

Update Farmer Profile

Farmer Access

Updates farm holding metrics, contact telephone number, district location, and avatar image URI.

HEADERS

Content-Type: application/json | Authorization: Bearer

REQUEST BODY (JSON)

```
{
  "email": "aswin1210@gmail.com",
  "name": "ASHWIN K S",
  "phone": "9345675687",
  "district": "Wayanad",
  "crop": "rice (basmathi)",
  "acres": 5.5,
  "soil_type": "Loamy Alluvial"
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Farmer profile updated successfully in MySQL database"
}
```

3. Farmer Crop & Field Management Services

Enables farmers to log crop rotations, track field parcel acreage, monitor vegetative growth stages, and assess crop health scores.

GET

/api/user/crops

Retrieve Farmer Crops & Parcels

Farmer / Admin Access

Fetches list of active crops registered under the specified farmer ID or email.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
email	string	Optional	Farmer's registered email
farmer_id	string	Optional	Unique farmer UUID

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "add_1790006014189_vgsn",
      "crop_name": "rice",
      "variety": "basmathi",
      "acres": 3,
      "health": 90,
      "growth_stage": "Growing",
      "status": "Available"
    },
    {
      "id": "add_1790005832080_w54k",
      "crop_name": "corn",
      "variety": "Hybrid / Standard Variety",
      "acres": 2.5,
      "health": 90,
      "growth_stage": "Growing",
      "status": "Available"
    }
  ]
}
```

POST /api/user/crops

Add New Crop Parcel

Farmer Access

Creates a new agricultural crop parcel in the MySQL `products` and `farmer\_crops` tables.

HEADERS

Content-Type: application/json | Authorization: Bearer

REQUEST BODY (JSON)

```
{
  "farmer_id": "u_1786594685785_3j73",
  "farmer_email": "aswin1210@gmail.com",
  "crop_name": "Pepper",
  "variety": "Panniyur-1",
  "quantity_acres": 1.5,
  "price_per_unit": 680,
  "health_rating": 95,
  "growth_stage": "Vegetative",
  "location_district": "Palakkad"
}
```

RESPONSE EXAMPLE [HTTP 201]

```
{
  "success": true,
  "message": "Crop parcel added successfully to farm inventory",
  "cropId": "crop_1790321550_9941"
}
```

DELETE /api/user/crops/:id

Delete Crop Parcel

Farmer / Admin Access

Removes a crop parcel record from the database by its unique parcel ID.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
id	string	Required	Crop parcel ID (path param)

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Crop parcel removed from database"
}
```

GET

/api/user/yield-trend

Farmer Yield & Harvest Analytics

Farmer Access

Calculates harvest progression across crop parcels, production totals in Quintals, and estimated revenue value.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
email	string	Required	Farmer email

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": {
    "ytd_production_qtl": 123,
    "target_reached_pct": 90,
    "estimated_value_inr": 161000,
    "timeline": [
      {
        "week": "Week 1",
        "yield": 35
      },
      {
        "week": "Week 2",
        "yield": 70
      },
      {
        "week": "Week 3",
        "yield": 95
      },
      {
        "week": "Week 4",
        "yield": 123
      }
    ]
  }
}
```

4. Real-Time Push Notifications & Broadcast Center Services

Broadcasts critical agro-advisories, mandi price surge warnings, and government schemes directly to all farmers or targeted users in real-time.

POST /api/admin/notifications

Send Admin Broadcast Notification

Admin Access

Dispatches a push notification stored in MySQL `admin\_notifications`. Supports broadcasting to ALL farmers or targeting a SPECIFIED farmer, district, or crop in real-time.

HEADERS

Content-Type: application/json | Authorization: Bearer

REQUEST BODY (JSON)

```
{
  "title": "Monsoon Crop Drainage Advisory",
  "message": "Heavy showers predicted in Wayanad & Palakkad. Clear secondary drainage channels immediately.",
  "category": "Weather",
  "priority": "Urgent",
  "target_audience": "user",
  "target_value": "aswin1210@gmail.com",
  "sender_admin": "Admin Administrator"
}
```

RESPONSE EXAMPLE [HTTP 201]

```
{
  "success": true,
  "message": "Notification broadcasted to farmers successfully!",
  "data": {
    "id": "notif_1790320657_mdhh",
    "title": "Monsoon Crop Drainage Advisory",
    "message": "Heavy showers predicted in Wayanad & Palakkad.",
    "category": "Weather",
    "priority": "Urgent",
    "target_audience": "user",
    "target_value": "aswin1210@gmail.com",
    "sender_admin": "Admin Administrator",
    "status": "active",
    "created_at": "2026-09-25T07:17:37.000Z"
  }
}
```

GET /api/notifications

Farmer Live Notification Feed

Farmer Access

Fetches live notifications in real-time. Automatically filters records matching 'all', or matching the farmer's specific email, user ID, district, or crop.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
user_email	string	Optional	Farmer's registered email
user_id	string	Optional	Farmer ID
district	string	Optional	Farmer district
crop	string	Optional	Farmer crop

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "notif_1790320657_mdhh",
      "title": "Monsoon Crop Drainage Advisory",
      "message": "Heavy showers predicted in Wayanad & Palakkad.",
      "category": "Weather",
      "priority": "Urgent",
      "target_audience": "user",
      "target_value": "aswin1210@gmail.com",
      "sender_admin": "Admin Administrator",
      "created_at": "2026-09-25T07:17:37.000Z"
    }
  ]
}
```

GET /api/admin/notifications

List All Broadcast History

Admin Access

Retrieves complete log of all broadcast alerts dispatched across Kerala.

HEADERS

Authorization: Bearer

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "notif_1790319063676_mqg6",
      "title": "heavy rain fall",
      "message": "Advisory for coastal districts",
      "category": "Market",
      "priority": "Urgent",
      "target_audience": "district",
      "target_value": "Kannur",
      "created_at": "2026-09-25T06:51:03.000Z"
    }
  ]
}
```

DELETE /api/admin/notifications/:id

Delete Notification Broadcast

Admin Access

Permanently deletes a broadcast record from MySQL database.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
id	string	Required	Notification ID

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Notification broadcast deleted successfully."
}
```

5. Mandi Market Price Intelligence Services

Monitors daily wholesale Mandi trading prices, modal rates, price shifts, and AI price forecasting across Kerala agricultural markets.

GET

/api/market-prices

Farmer Mandi Commodity Rates

Public / Farmer Access

Returns wholesale APMC Mandi rates with minimum, maximum, and modal price indices per commodity.

HEADERS

None required

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
district	string	Optional	Filter by district (e.g. Palakkad, Wayanad)
crop	string	Optional	Filter by commodity name

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "m_paddy_real",
      "crop_name": "Organic Paddy (Jyothi)",
      "district": "Palakkad",
      "min_price": 2180,
      "max_price": 2350,
      "modal_price": 2280,
      "unit": "Quintal",
      "price_trend": "+3.3%",
      "status": "Rising"
    },
    {
      "id": "m_pepper_real",
      "crop_name": "Pepper",
      "district": "Palakkad",
      "min_price": 640,
      "max_price": 720,
      "modal_price": 680,
      "unit": "KG",
      "price_trend": "+6.3%",
      "status": "Rising"
    }
  ]
}
```

POST /api/admin/market-prices

Create Mandi Price Entry

Admin Access

Adds a new commodity trading price quote for a designated Mandi.

HEADERS

Content-Type: application/json | Authorization: Bearer

REQUEST BODY (JSON)

```
{
  "crop_name": "Cardamom (Green Supreme)",
  "district": "Idukki",
  "min_price": 1850,
  "max_price": 2150,
  "modal_price": 2000,
  "unit": "KG",
  "price_trend": "+5.1%",
  "status": "Rising"
}
```

RESPONSE EXAMPLE [HTTP 201]

```
{
  "success": true,
  "message": "Market price record published successfully"
}
```

POST /api/admin/market-prices/upsert

Bulk Upsert Mandi Rates

Admin Access

Batch synchronizes multiple commodity market rates in a single atomic transaction.

HEADERS

Content-Type: application/json | Authorization: Bearer

REQUEST BODY (JSON)

```
{
  "prices": [
    {
      "crop": "Paddy",
      "current": 2280,
      "prev": 2180,
      "trend": "+4.5%"
    },
    {
      "crop": "Corn",
      "current": 1680,
      "prev": 1620,
      "trend": "+3.7%"
    }
  ]
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Bulk market rates updated successfully"
}
```

6. Weather & Agro-Climatic Intelligence Services

Hyper-local weather telemetry delivering precipitation forecasts, humidity readings, solar radiation indices, and farm management advice.

GET

/api/weather

Get Localized Agro-Weather Forecast

Public / Farmer Access

Fetches 7-day meteorological forecasts, precipitation alerts, and fieldwork safety advisories.

HEADERS

None required

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
district	string	Optional	District name (default: Palakkad)

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": {
    "district": "Palakkad",
    "current": {
      "temp_c": 28.5,
      "condition": "Sunny with light clouds",
      "humidity": 78,
      "wind_kph": 12.4,
      "rain_chance_pct": 20
    },
    "forecast": [
      {
        "day": "Today",
        "high": 31,
        "low": 24,
        "rain": 20,
        "desc": "Sunny"
      },
      {
        "day": "Tomorrow",
        "high": 29,
        "low": 23,
        "rain": 65,
        "desc": "Rain"
      }
    ],
    "advisory": "Favorable conditions for foliar fertilizer application before midday."
  }
}
```

7. AI Diagnostics & Ollama LLM Services

Deep learning models for plant pathology detection, leaf disease classification, and multi-turn LLM agricultural advisory via local Ollama instances.

GET /api/ai/ollama/status

Ollama LLM Engine Connectivity Check

Public Access

Checks local Ollama server status (`http://localhost:11434`), loaded neural models, and inference availability.

HEADERS

None required

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "connected": true,
  "model": "llama3",
  "version": "0.1.32",
  "modelsAvailable": [
    "llama3:latest",
    "mistral:latest"
  ]
}
```

POST /api/ai/chat

Conversational Farming Assistant

Farmer Access

Multi-turn AI conversation answering questions on soil management, fertilizer formulas, pest control, and market timing.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "message": "What fertilizer should I apply for my rice paddy at tillering stage?",
  "district": "Palakkad",
  "crop": "Paddy"
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "reply": "For rice at tillering stage, apply Urea (46-0-0) at 50 kg/ha as top-dressing. Ensure soil moisture is adequate.",
  "modelUsed": "llama3"
}
```

POST /api/crops/scan

Crop Leaf Disease Image Diagnostic

Farmer Access

Analyzes crop leaf symptoms via computer vision/Ollama to detect diseases, pathogen type, severity level, and treatments.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "farmer_id": "u_1790170235463_3crv",
  "crop": "Paddy",
  "district": "Palakkad",
  "image_data": "..."
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "scanId": "scan_1790322100",
  "disease": "Bacterial Blight (Xanthomonas oryzae)",
  "severity": "Medium",
  "confidence": 96,
  "treatment": "Apply Streptomycin sulphate + Tetracycline combination with Copper Oxychloride. Drain standi",
  "status": "detected"
}
```

GET

/api/crops/scans

Farmer Disease Scan History

Farmer Access

Retrieves historical disease diagnostic scans conducted by the authenticated farmer.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
farmer_id	string	Required	Farmer UUID

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "scan_1790093429813",
      "crop": "paddy",
      "disease": "Bacterial Blight",
      "severity": "Medium",
      "confidence": 96,
      "status": "detected",
      "detected_at": "2026-09-22T16:10:29.000Z"
    }
  ]
}
```

8. Admin Oversight & Analytics Services

Administrative intelligence center for farmer verification, regional crop census, account governance, and platform KPIs.

GET /api/admin/overview

Executive Metrics & Regional KPI Rollup

Admin Access

Returns top-level management KPIs: total registered farmers, active percentage, aggregated crop acreage, and district distributions.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
district	string	Optional	Filter by district

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": {
    "totalFarmers": 5,
    "activeFarmers": 5,
    "totalAdmins": 1,
    "totalCrops": 5,
    "totalCropAcres": "11.0 ac",
    "diseaseResolutionRate": "98%",
    "totalMandis": "28 Mandis",
    "cropDistribution": [
      {
        "name": "Paddy (Jyothi)",
        "count": 2,
        "acres": "3.5",
        "value": 32
      },
      {
        "name": "rice (basmathi)",
        "count": 1,
        "acres": "3.0",
        "value": 27
      },
      {
        "name": "corn",
        "count": 1,
        "acres": "2.5",
        "value": 23
      },
      {
        "name": "Pepper",
        "count": 1,
        "acres": "1.0",
        "value": 9
      },
      {
        "name": "Rubber (RSI 4)",
        "count": 1,
        "acres": "1.0",
        "value": 9
      }
    ]
  }
}
```

```
}  
}
```

GET /api/admin/farmers

Admin Farmer Registry Query

Admin Access

Paginated search across the master farmer database with filtering by status and query string.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
page	number	Optional	Page index (default: 1)
limit	number	Optional	Records per page (default: 20)
search	string	Optional	Keyword search in name, email, crop, location
status	string	Optional	Filter by status: 'active', 'inactive', or 'all'

RESPONSE EXAMPLE [HTTP 200]

```
{  
  "success": true,  
  "count": 5,  
  "data": [  
    {  
      "id": "u_1790170235463_3crv",  
      "name": "thilaga",  
      "email": "717824f155@gmail.com",  
      "phone": "6380514411",  
      "district": "Palakkad",  
      "crop": "Pepper",  
      "acres": 1,  
      "status": "Active"  
    },  
    {  
      "id": "u_farmer_01",  
      "name": "Ramesh Kumar",  
      "email": "ramesh@gmail.com",  
      "phone": "+91 94470 12345",  
      "district": "Palakkad",  
      "crop": "Organic Paddy (Jyothi Hybrid)",  
      "acres": 2.5,  
      "status": "Active"  
    }  
  ]  
}
```

PUT /api/admin/farmers/:id/status

Update Farmer Account Status

Admin Access

Modifies farmer status (e.g. Active, Suspended, Inactive).

HEADERS

Content-Type: application/json | Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
id	string	Required	Farmer user ID

REQUEST BODY (JSON)

```
{
  "status": "Active"
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Farmer status updated to Active"
}
```

GET

/api/admin/analytics

Growth & Telemetry Analytics

Admin Access

Provides time-series growth telemetry on sessions, farmer onboarding, and revenue.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
month	string	Optional	Specific month or 'all'
district	string	Optional	District filter

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "hasData": true,
  "data": [
    {
      "month": "May",
      "farmers": 1,
      "sessions": 45,
      "revenue": 1800
    },
    {
      "month": "Aug",
      "farmers": 4,
      "sessions": 280,
      "revenue": 11200
    },
    {
      "month": "Sep",
      "farmers": 5,
      "sessions": 420,
      "revenue": 16800
    }
  ],
  "meta": {
    "totalFarmers": 5,
    "totalSessions": 420,
    "totalRevenue": 16800
  }
}
```

9. Public Support & Inquiries Services

Public contact form processing, automated admin alert routing, and support ticket management.

POST /api/contact

Submit Public Contact Message

Public Access

Accepts inquiry from landing page contact form. Saves to `contact\_messages` table and immediately raises an alert in the Admin Portal.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "name": "Dr. Ananya Nair",
  "email": "ananya@agriuniversity.edu",
  "subject": "Partnership & Soil Sensor Research",
  "message": "We would like to integrate our soil moisture telemetry with FARMO AI."
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "Thank you! Your message has been submitted to the admin team.",
  "data": {
    "id": "contact_1790322500_xyz1",
    "name": "Dr. Ananya Nair",
    "email": "ananya@agriuniversity.edu",
    "subject": "Partnership & Soil Sensor Research",
    "created_at": "2026-09-25T08:15:00.000Z"
  }
}
```

GET /api/admin/contact-messages

List Contact Inquiries

Admin Access

Fetches submitted user inquiries, sorted chronologically with read/unread statuses.

HEADERS

Authorization: Bearer

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "data": [
    {
      "id": "contact_1790322500_xyz1",
      "name": "Dr. Ananya Nair",
      "email": "ananya@agriuniversity.edu",
      "subject": "Partnership & Soil Sensor Research",
      "message": "We would like to integrate our soil moisture telemetry with FARMO AI.",
      "status": "unread",
      "created_at": "2026-09-25T08:15:00.000Z"
    }
  ]
}
```

PUT

/api/admin/contact-messages/:id/read

Mark Contact Message Read

Admin Access

Updates message status to 'read'.

HEADERS

Authorization: Bearer

QUERY / PATH PARAMETERS

Parameter	Type	Required	Description
id	string	Required	Contact message ID

RESPONSE EXAMPLE [HTTP 200]

```
{  "success": true,  "message": "Message marked as read"}
```

10. Reports & Data Export Services

Standardized tabular reporting, agricultural production census, and unified data views.

GET /api/reports/users

Master Users & Farmers Census Report

Admin Access

Produces full tabular dump combining user accounts, farmer credentials, crop varieties, farm land acreage, and registered dates.

HEADERS

Authorization: Bearer

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "count": 6,
  "data": [
    {
      "user_id": "u_admin_default",
      "user_name": "Admin Administrator",
      "email": "admin@gmail.com",
      "role": "admin",
      "district": "Kerala"
    },
    {
      "user_id": "u_1786594685785_3j73",
      "user_name": "ASHWIN",
      "email": "aswin1210@gmail.com",
      "role": "farmer",
      "district": "Wayanad",
      "crop_name": "rice (basmathi), corn",
      "crop_acres": 5.5
    }
  ]
}
```

GET /api/unified/all

Unified All-in-One Data View

Admin Access

High-performance SQL View (`view\_unified\_all\_in\_one6`) consolidating users, farmer crops, disease scans, and market prices.

HEADERS

Authorization: Bearer

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "totalRows": 6,
  "viewName": "view_unified_all_in_one6"
}
```

11. Spring Boot Java Microservices Architecture

Enterprise Java Spring Boot microservice cluster running on port 8080 (`com.farmoai.microservices`), providing modular JPA-backed endpoints.

POST /api/auth/register (Java Spring Boot)

Spring Boot User Registration Microservice

Public Access

Spring Data JPA controller handling farmer account registration with duplicate email validation and password encryption.

HEADERS

Content-Type: application/json

REQUEST BODY (JSON)

```
{
  "name": "Manoj Varma",
  "email": "manoj@gmail.com",
  "district": "Thrissur",
  "phone": "+91 94470 55555"
}
```

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "message": "User registered successfully via Spring Boot Microservice"
}
```

GET /api/crops/all (Java Spring Boot)

Spring Boot Crop Management Microservice

Farmer Access

JPA repository-backed crop entity retrieval with health rating filtering.

HEADERS

Authorization: Bearer

RESPONSE EXAMPLE [HTTP 200]

```
{
  "success": true,
  "count": 5,
  "service": "CropManagementMicroservice"
}
```

12. Data Models, Schemas & Entity Dictionary

Database table structures across the MySQL relational schema `farmo\_ai\_db`.

Relational Database Schemas (MySQL farmo_ai_db)

Normalized table definitions and entity schemas residing in the MySQL database.

Table Name	Columns & Relational Model Description
users	Core user authentication table (id, full_name, email, password, phone, role, district, created_at)
farmers	Agricultural farmer profiles (id, full_name, email, phone, location, district, primary_crop, farm_size_acres, status)
products	Crops and farm harvest parcels (id, farmer_id, product_name, variety, quantity_acres, price_per_unit, health_rating, growth_stage)
farmer_crops	Crop rotation records and field tracking (id, farmer_id, crop_name, variety, acres, planting_date, harvest_date, health)
admin_notifications	Push notification broadcasts (id, title, message, category, priority, target_audience, target_value, sender_admin, status, created_at)
contact_messages	Public user contact inquiries (id, name, email, subject, message, status, created_at)
disease_scans	Plant pathology scan records (id, farmer_id, crop, disease, severity, confidence, treatment, status, created_at)
market_prices	APMC wholesale Mandi price index (id, crop_name, district, min_price, max_price, modal_price, unit, price_trend, status)

Error Codes & Standard Response Structure

HTTP Code	Error Name	Description & Resolution
200 OK	Success	Request completed successfully and payload returned.
201 Created	Resource Created	New account, crop parcel, or broadcast notification created.
400 Bad Request	Validation Error	Missing required fields (e.g. email, password, notification title).
401 Unauthorized	Auth Required	Missing or expired Bearer token in request header.
403 Forbidden	Role Permission Denied	Attempting to access Admin operations without superuser privileges.
404 Not Found	Resource Missing	Target entity ID (farmer, crop, notification) not found in database.
500 Server Error	Internal Error	Database timeout, connection disconnect, or syntax exception.